

Fundamentals of Linear Control:

A Concise Approach

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Chapter 2: Dynamic Systems

linearcontrol.info/fundamentals

2.1 Dynamic Models

Longitudinal car model

Newton's law

$$m \ddot{x} + b \dot{x} = f(t)$$

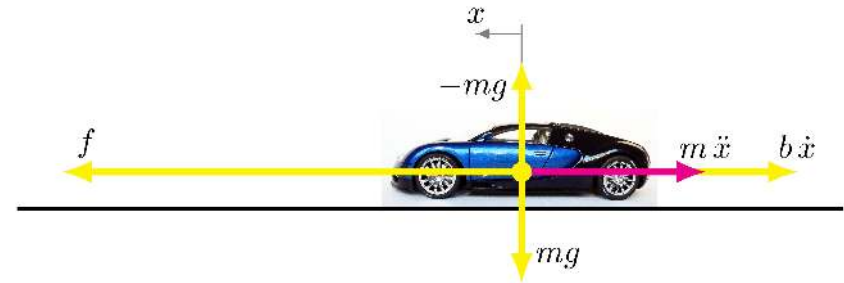
Force model

$$f(t) = p u(t)$$

Ordinary differential equation

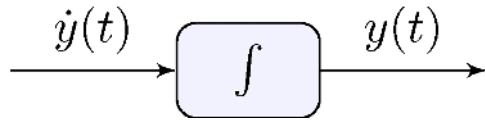
$$\ddot{y} + \frac{b}{m} \dot{y} = \frac{p}{m} u(t)$$

- input: u is pedal excursion in inches
- output: y is car velocity in mph

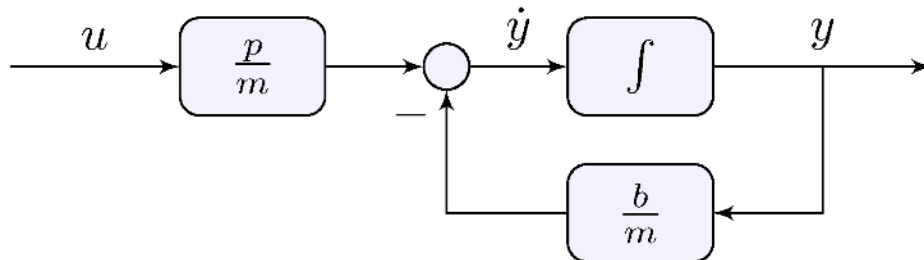


2.2 Block-Diagrams for Differential Equations

Basic element: integrator



Longitudinal car model



$$\dot{y}(t) = \frac{p}{m} u(t) - \frac{b}{m} y(t)$$

Note the presence of feedback!

2.3 Dynamic Response

Longitudinal car model

$$\begin{aligned} \dot{y}(t) + \frac{b}{m} y(t) &= \frac{p}{m} u(t) \end{aligned}$$

Constant pedal input

$$u(t) = \tilde{u}, \quad t \geq 0$$

Particular solution

$$y_P(t) = \tilde{y} = \left(\frac{p}{m} \middle/ \frac{b}{m} \right) \tilde{u} = \frac{p}{b} \tilde{u}$$

Longitudinal car model: homogenous equation

$$\dot{y}(t) + \frac{b}{m} y(t) = 0$$

Homogenous solution

$$y_H(t) = e^{\lambda t}$$

Characteristic equation

$$\dot{y}_H(t) + \frac{b}{m} y_H(t) = \left(\lambda + \frac{b}{m} \right) e^{\lambda t} = 0 \quad \implies \quad \lambda + \frac{b}{m} = 0$$

Longitudinal car model with constant pedal input

$$\begin{aligned} \dot{y}(t) + \frac{b}{m} y(t) &= \frac{p}{m} \tilde{u} \end{aligned}$$

Particular + homogenous solution

$$\begin{aligned} y(t) &= y_P(t) + \beta, \quad y_H(t) = \tilde{y} + \beta e^{\lambda t} \end{aligned}$$

Initial conditions

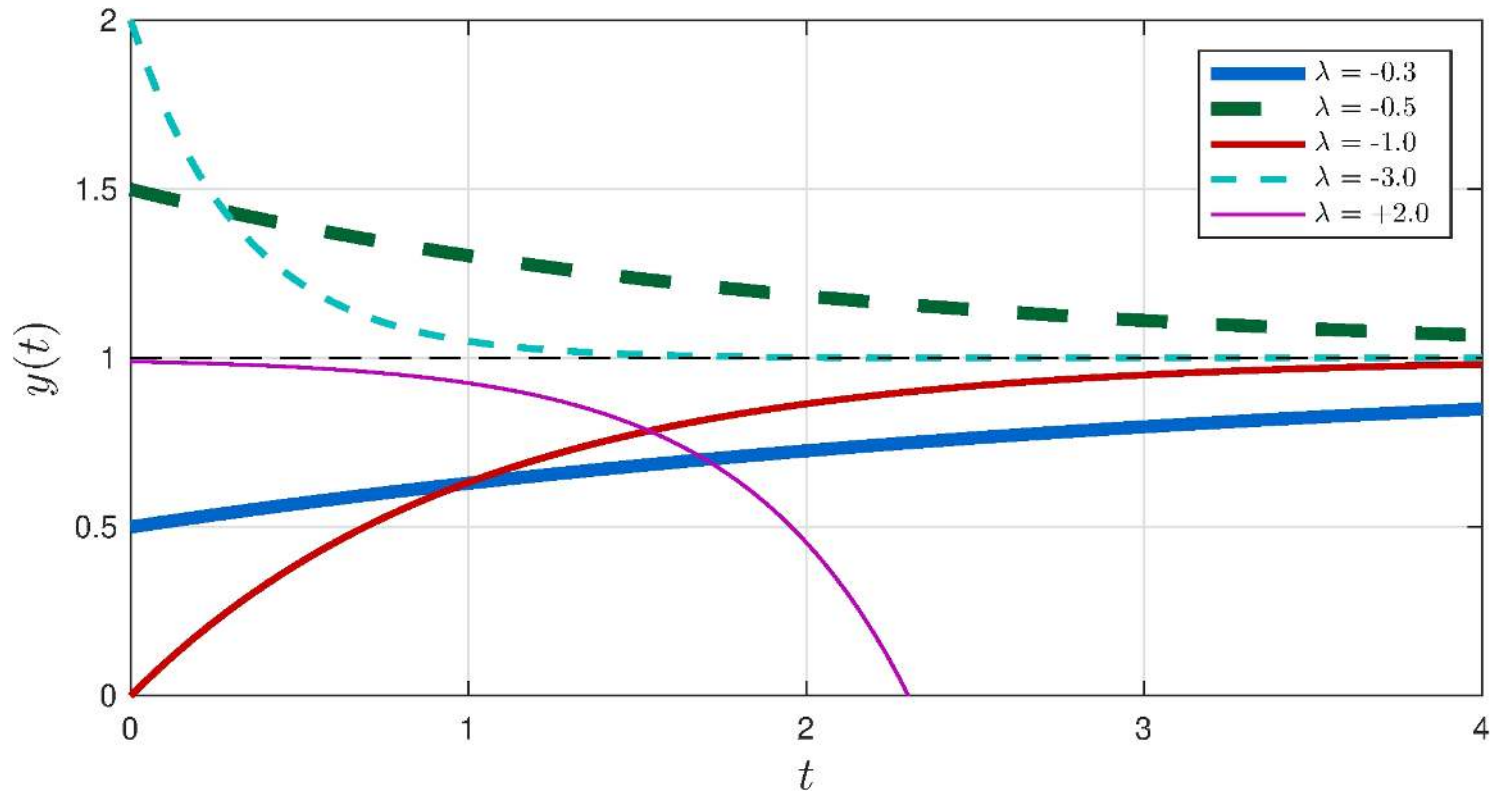
$$\begin{aligned} y(0) = \tilde{y} + \beta &= y_0 \quad \& \implies \quad \beta = y_0 - \tilde{y} \end{aligned}$$

Complete solution

$$\begin{aligned} y(t) &= \tilde{y} \left(1 - e^{\lambda t} \right) + y_0 e^{\lambda t}, \quad \lambda = -\frac{b}{m}, \quad \tilde{y} = \frac{p}{b} \tilde{u} \end{aligned}$$

Examples

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\[ \begin{align*} y(t) &= \tilde{y} \left( 1 - e^{\lambda t} \right) + y_0 e^{\lambda t}, & \tilde{y} &= 1 \\ \end{align*} \]
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Time-constant

Definition

$$\tau = \left\{ t : y(t) = \left(1 - e^{-1} \right) \tilde{y} \approx 0.63 \tilde{y} \right\}$$

First-order systems

$$\tau = -\frac{1}{\lambda} \quad y(\tau) \approx 0.63 \tilde{y}, \quad y(3\tau) \approx 0.95 \tilde{y}$$

Rise-time

Definition

$$t_r = t_2 - t_1, \quad t_1 = \left\{ t : y(t) = 0.1 \tilde{y} \right\}, \quad t_2 = \left\{ t : y(t) = 0.9 \tilde{y} \right\}$$

First-order systems

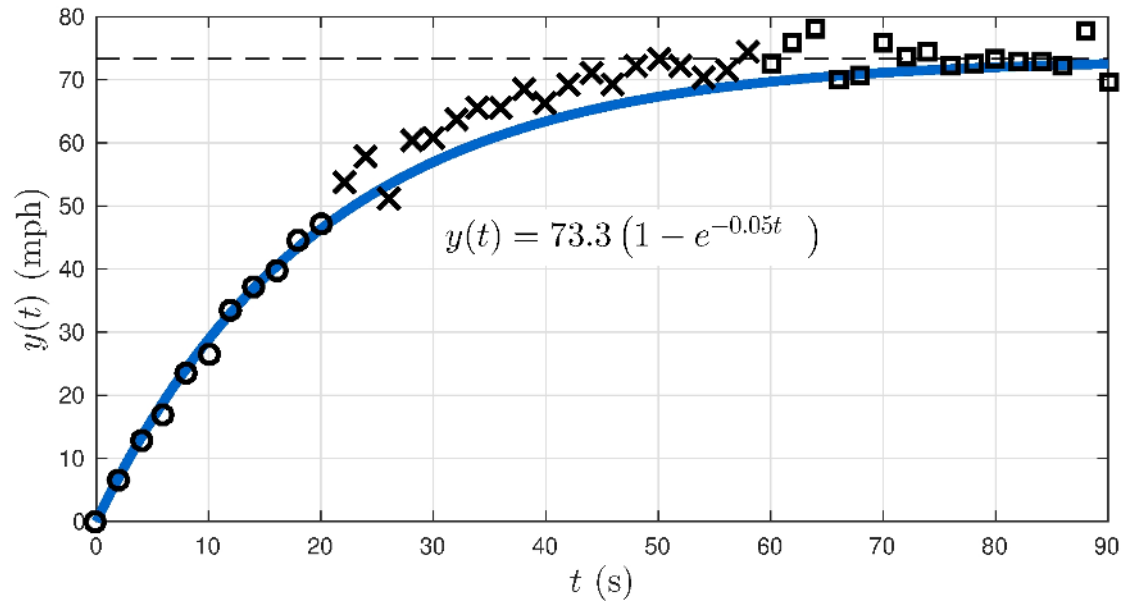
$$t_r = \ln\left(\frac{1}{9} \right) \cdot \lambda^{-1} = \ln(9) \tau \approx 2.2 \tau$$

2.4 Experimental Dynamic Response

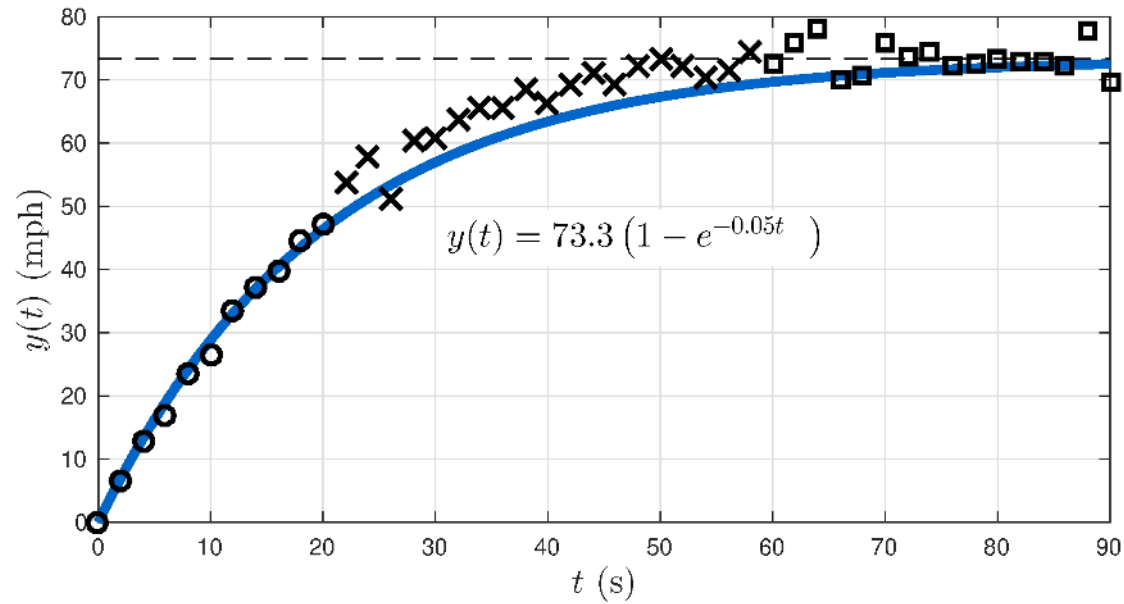
Experiment

Zero initial velocity

Constant pedal excursion: $\tilde{u} = 1$ in



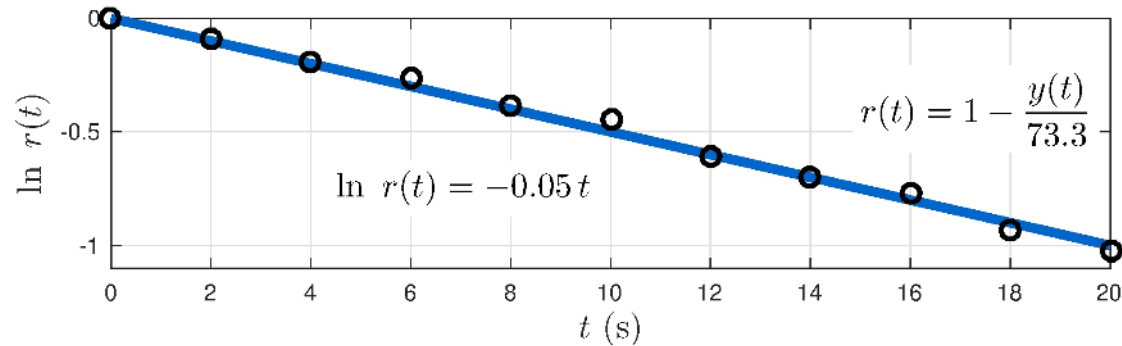
Model fitting



Squares: steady-state solution $\implies \tilde{y} \approx 73.3 \text{ mph}$

Model fitting

$$\begin{aligned} r(t) &= 1 - \frac{y(t)}{\tilde{y}} \quad . = e^{\lambda t} \quad \& \quad \implies \quad \ln r(t) = \lambda t \\ \end{aligned}$$



Circles: $\lambda \approx -0.05 \text{ s}^{-1}$

Model fitting

$$\begin{aligned} \frac{b}{m} &= -\lambda \approx 0.05 \text{ s}^{-1}, \quad \frac{p}{b} = \frac{\tilde{y}}{\tilde{u}} \\ &\approx 73.3 \text{ mph/in}, \quad \frac{p}{m} = \frac{b}{m} \times \frac{p}{b} = -\lambda \times \frac{\tilde{y}}{\tilde{u}} \\ &= 3.7 \text{ mph/(in s)} \end{aligned}$$

Experimental model

$$\dot{y}(t) + 0.05 y(t) = 3.7 u(t)$$

Be careful

Experiment was conducted for $u(t) = \tilde{u}$ constant but we *wish* model to hold for general $u(t)$!

2.5 Dynamic Feedback Control

Car model

Dynamic model

$$\begin{aligned} \dot{y}(t) + \frac{b}{m} y(t) &= \frac{p}{m} u(t) \end{aligned}$$

Feedback controller

$$\begin{aligned} u(t) &= K (\bar{y} - y(t)) \end{aligned}$$

Closed-loop system

$$\begin{aligned} \dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) &= \frac{p}{m} K \bar{y} \end{aligned}$$

Closed-loop response

Dynamic model

$$\begin{aligned} \dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) &= \frac{p}{m} K \bar{y}, & y(0) &= y_0, & \lambda &= -\frac{b}{m} - \frac{p}{m} K \end{aligned}$$

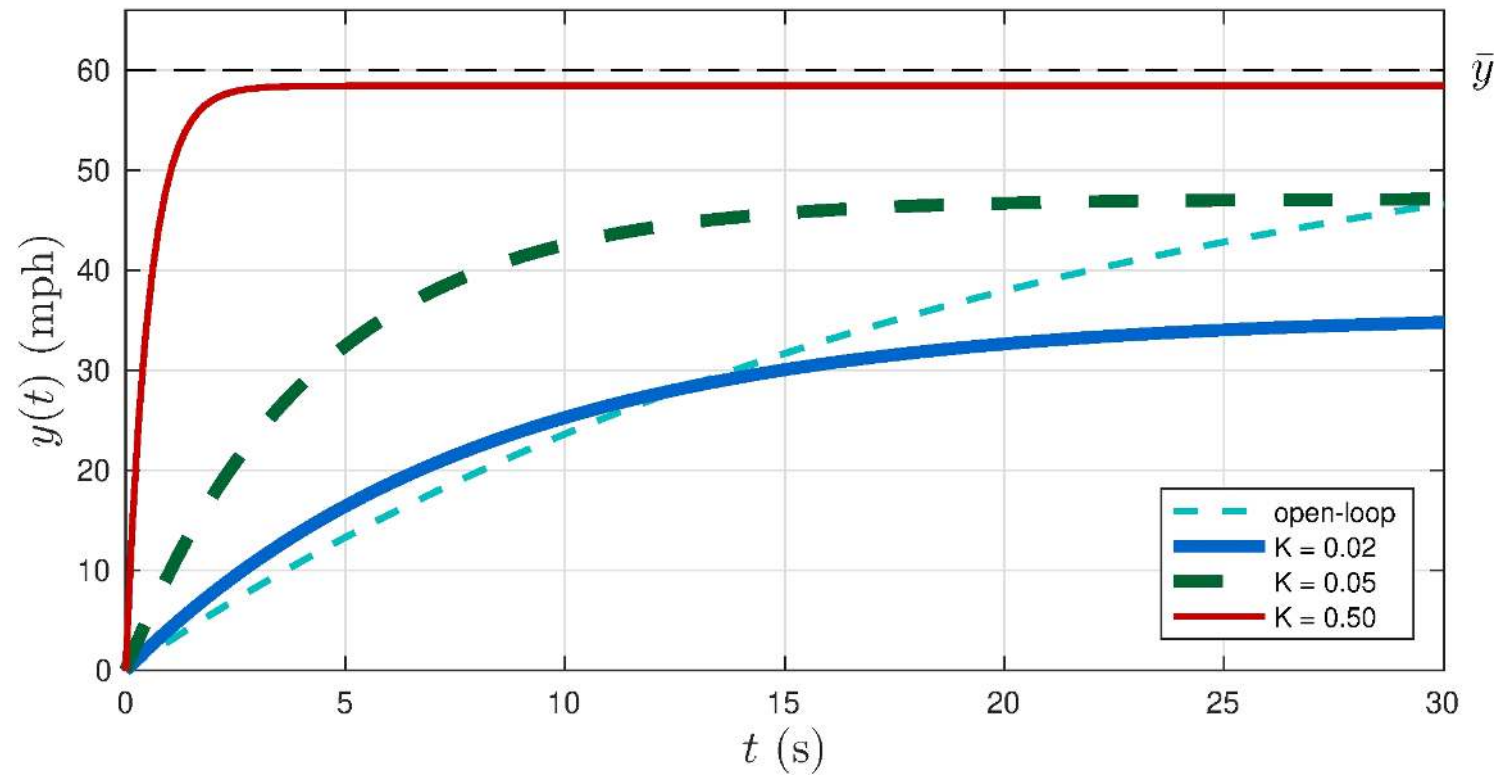
Output

$$\begin{aligned} y(t) &= \tilde{y} \left(1 - e^{\lambda t} \right) + y_0 e^{\lambda t}, & \tilde{y} &= \frac{\frac{p}{m} K}{\frac{b}{m} + \frac{p}{m} K} \bar{y} = \frac{\frac{p}{m} K}{1 + \frac{p}{b} K} \bar{y} \end{aligned}$$

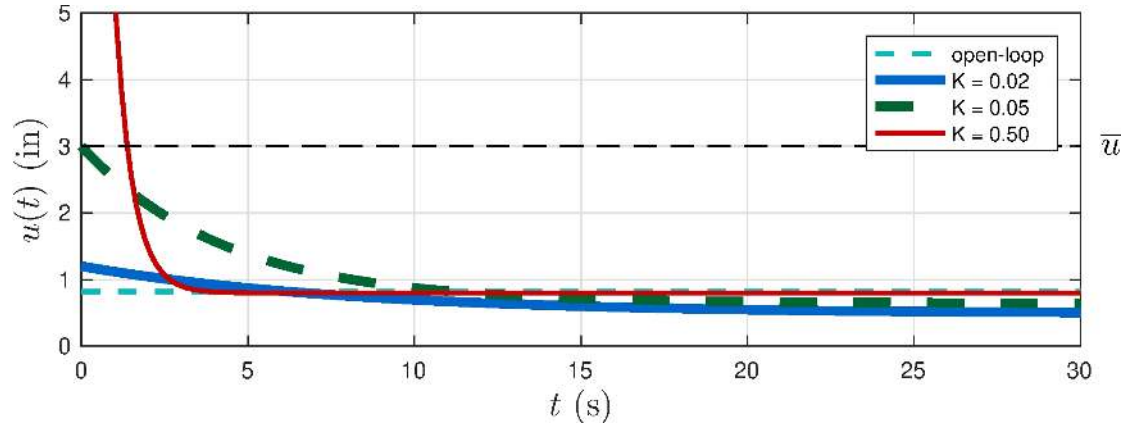
Time-constant

$$\begin{aligned} \lambda &= -\frac{b}{m} - \frac{p}{m} K, & \tau &= -\lambda^{-1} = \frac{m}{b + p K} \end{aligned}$$

Output



Control



Largest control at $(t = 0)$

$$\begin{aligned} u(0) &= K e(0) = K (\bar{y} - y(0)) = K \bar{y} \end{aligned}$$

Saturation

$$\bar{y} = 60 \text{ mph} \implies u(t) > \bar{u} = 3 \text{ in for } K > 0.05!$$

Steady-state

$$\begin{aligned} \lim_{t \rightarrow \infty} y(t) &= \tilde{y} = H(0) \bar{y}, & H(0) &= \frac{\frac{p}{b} K}{1 + \frac{p}{b} K} \\ \lim_{t \rightarrow \infty} e(t) &= \bar{y} - \tilde{y} = S(0) \bar{y}, & S(0) &= \frac{1}{1 + \frac{p}{b} K} \end{aligned}$$

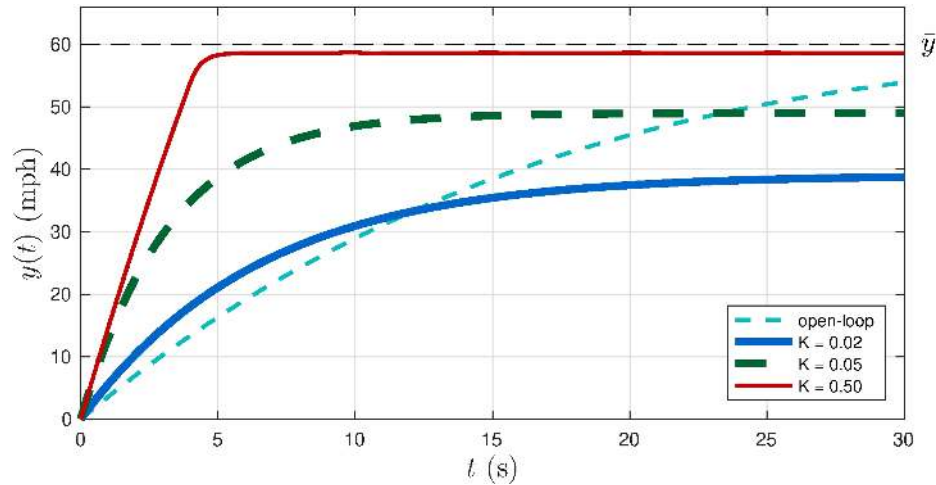
Effect of the gain

K	$H(0)$	$S(0)$	$\tilde{y}$ (mph)	τ (s)	t_r (s)
open-loop	1.00	0.00	60	20.0	43.9
0.02	0.60	0.40	36	8.1	17.8
0.05	0.79	0.21	47	4.3	9.4
0.50	0.97	0.03	58	0.5	1.2

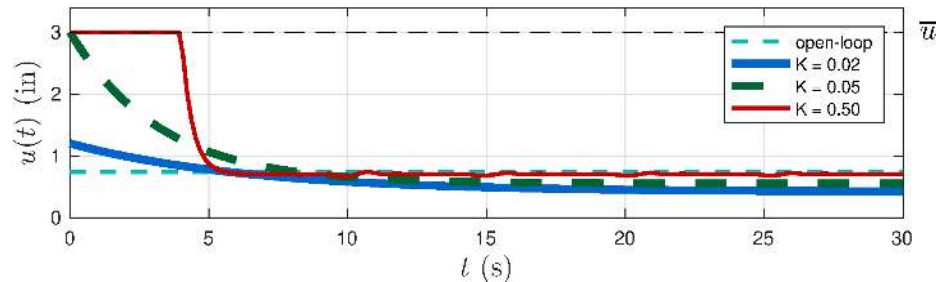
$$\bar{y} = 60 \text{ mph}$$

2.6 Nonlinear Models

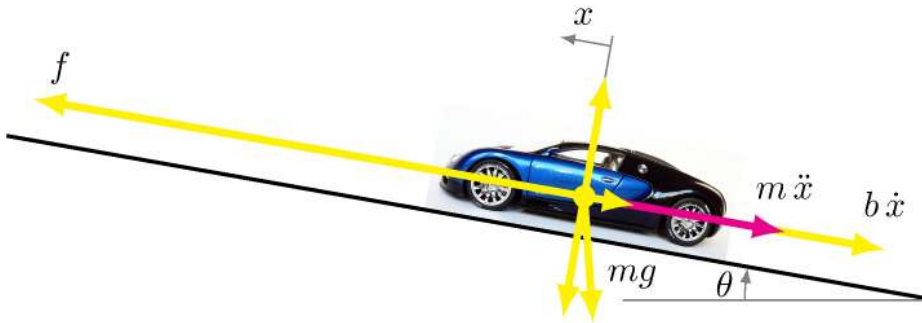
Output is rate-limited



Control saturates



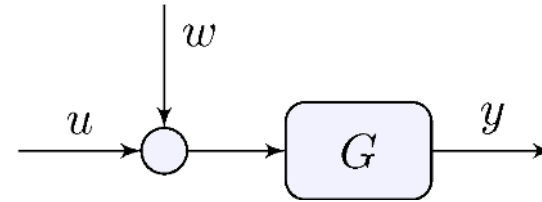
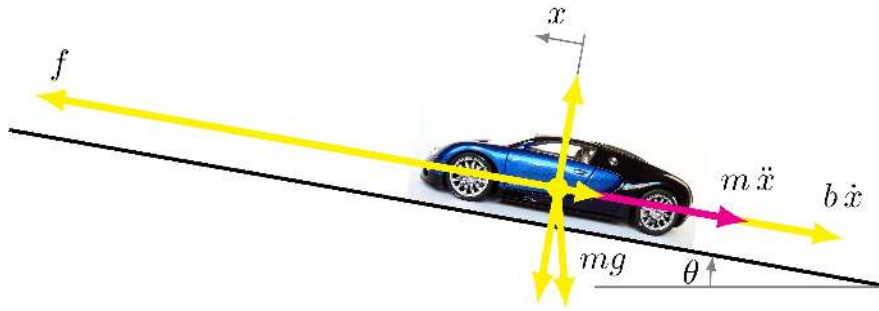
2.7 Disturbance Rejection



Car on inclined plane model

$$\ddot{y}(t) + \frac{b}{m} \dot{y}(t) = \frac{p}{m} u(t) - g \sin(\theta(t))$$

Disturbance Model



Car on inclined plane with disturbance

$$\begin{aligned} \dot{y}(t) + \frac{b}{m} y(t) &= \frac{p}{m} (u(t) + w(t)), & w(t) &= -\frac{mg}{p} \sin(\theta(t)) \end{aligned}$$

Car with disturbance

$$\begin{aligned} \dot{y}(t) + \frac{b}{m} y(t) &= \frac{p}{m} (u(t) + w(t)) \end{aligned}$$

Constant disturbance

$$\begin{aligned} w(t) &= \bar{w}, \quad t \geq 0 \end{aligned}$$

Feedback controller

$$\begin{aligned} u(t) &= K e(t), \quad e(t) = \bar{y} - y(t) \end{aligned}$$

Impact of disturbance on steady-state

Model

$$\begin{aligned} \dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) &= \frac{p}{m} \left(K \bar{y} + \bar{w} \right) \\ y_0 &= H(0) \bar{y} \end{aligned}$$

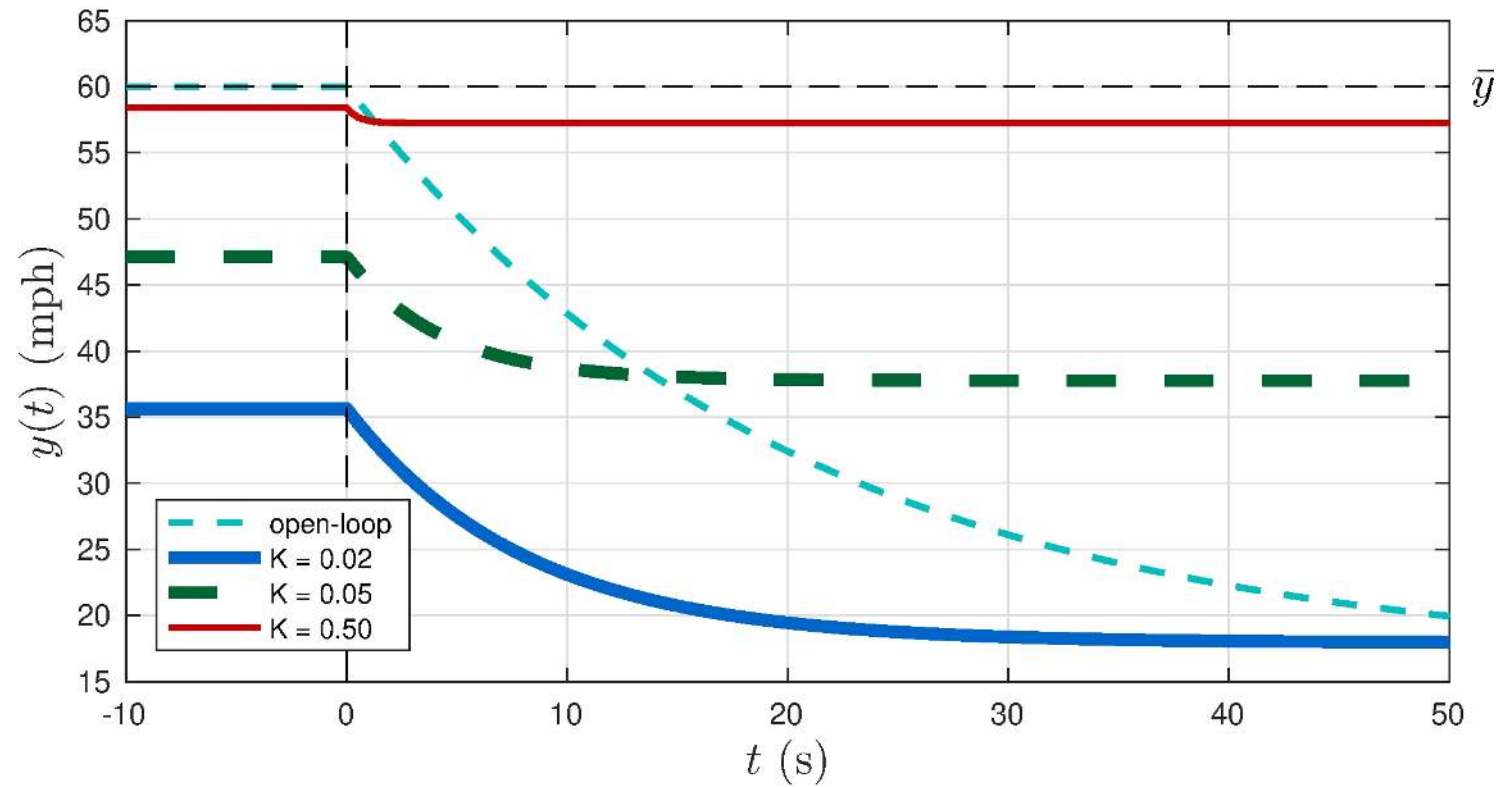
Output

$$\begin{aligned} y(t) &= H(0) \bar{y} + \left(1 - e^{-\lambda t} \right) D(0) \bar{w}, & H(0) &= \frac{\frac{p}{b}}{K \left(1 + \frac{p}{b} K \right)}, & D(0) &= \frac{p}{b} \frac{1}{1 + \frac{p}{b} K} \end{aligned}$$

Deviation from steady-state

$$\Delta y(t) = y(t) - H(0) \bar{y} = \left(1 - e^{-\lambda t} \right) D(0) \bar{w}$$

Output



2.8 Integral Action

Toilet tank model

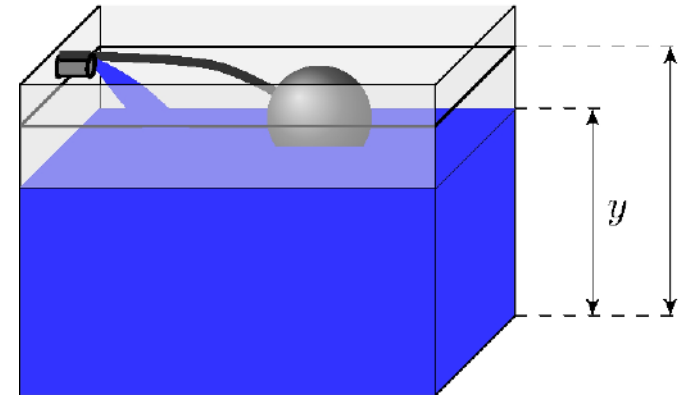
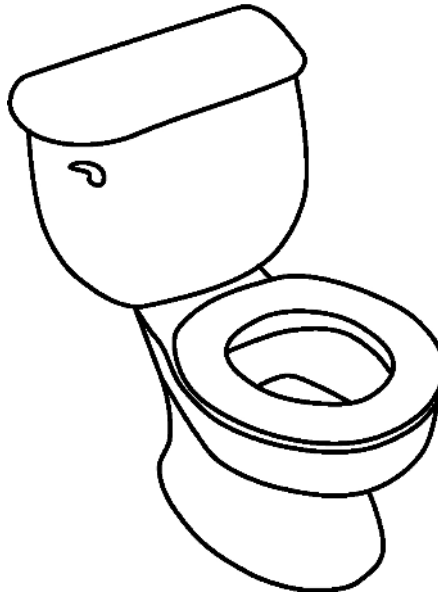
$$\begin{aligned} \dot{y}(t) &= \frac{1}{A} u(t) \end{aligned}$$

Ballcock valve

$$\begin{aligned} u(t) &= K(\bar{y} - y) \end{aligned}$$

Closed-loop model

$$\begin{aligned} \dot{y}(t) + \frac{K}{A} y(t) &= \frac{K}{A} \bar{y}(t) \end{aligned}$$



Closed-loop response

Model

$$\begin{aligned} \dot{y}(t) + \frac{K}{A} y(t) &= \frac{K}{A} \bar{y}(t), & y(0) &= y_0, & \lambda &= -\frac{K}{A} \\ \end{aligned}$$

Output

$$\begin{aligned} y(t) &= \tilde{y} \left(1 - e^{\lambda t} \right) + y_0 e^{\lambda t}, & \tilde{y} &= \bar{y} \\ \end{aligned}$$

Steady-state

$$\begin{aligned} \lim_{t \rightarrow \infty} y(t) &= \tilde{y} = H(0) \bar{y}, & H(0) &= 1 \\ \lim_{t \rightarrow \infty} e(t) &= \bar{y} - \tilde{y} = S(0) \bar{y}, & S(0) &= 0 \end{aligned}$$

Integral action $\implies$ zero tracking error!

Can we replicate that result in a more general setting? Answer is in Chapter 4

Learn more

- [Book website](#)

